
Technology Architecture Documentation

Integration between the sync2books Trader Invoicing System (TIS) and the KRA Electronic Tax Invoice Management System (eTIMS) via OSCU

Prepared for	Kenya Revenue Authority – eTIMS Integrator Certification
Product	sync2books — Trader Invoicing System (TIS)
Integration path	OSCU (Online Sales Control Unit)
Document version	1.0
Status	Target architecture — certification submission

1. Purpose and Scope

This document describes the technology architecture through which **sync2books** operates as a Trader Invoicing System (TIS) and integrates with the Kenya Revenue Authority (KRA) Electronic Tax Invoice Management System (eTIMS) as a certified third-party integrator, using the Online Sales Control Unit (OSCU) path.

sync2books connects a business's existing accounting platform (QuickBooks Online, Odoo, and other supported ERPs) to KRA eTIMS, so that sales, credit notes, stock movements, item catalogues and purchase records are exchanged with KRA in the sequence and format defined by the *TIS for OSCU/VSCU Technical Specifications v2.0*, without requiring the business to change how it invoices its customers.

This document is submitted in support of sync2books' application for eTIMS OSCU integrator certification, alongside the eTIMS Bio Data Form, Tax Compliance Certificate, and proof of qualified technical staff.

2. System Architecture Overview

The platform is composed of four cooperating services. Each has a single responsibility, communicates over authenticated HTTPS, and can be scaled or deployed independently.

Component	Responsibility
ERP connectors (QuickBooks Online, Odoo, MS Dynamics)	Source systems where the business creates invoices, credit notes, items, stock and purchase transactions. sync2books reads and writes to these via each platform's native API (OAuth2).
nest-sync-2-books-api (Core Sync Engine)	The system of record for tenant, connection, invoice and sync-job data. Normalises ERP entities into a common model, queues outbound eTIMS operations, and receives asynchronous compliance outcomes.
sync2books-compliance-api (eTIMS Compliance Service)	The only component that communicates directly with KRA eTIMS OSCU. Owns tenant/branch provisioning, device initialization, communication-key storage, request signing, and KRA response handling. Generates the KRA-signed receipt (PDF + QR code).
KRA eTIMS OSCU API	KRA's hosted Online Sales Control Unit. Validates submissions, assigns invoice/receipt numbers, and digitally signs each transaction.

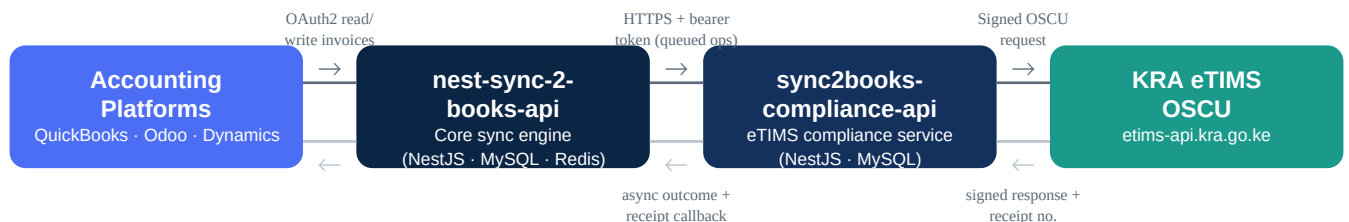


Figure 1. Component-level data flow between the accounting platform, the sync2books TIS, and KRA eTIMS OSCU.

2.1 Design principles

- **Single point of KRA contact.** Only the compliance service holds OSCU credentials (KRA PIN, branch office ID, device serial, communication key) and signs requests — the core sync engine and ERP connectors never call KRA directly.
- **Per-branch isolation.** Each customer branch is provisioned as its own OSCU device/connection (etims_branches table: KRA PIN, bhfid, device serial, connection status), matching KRA's per-branch initialization model.
- **Asynchronous, queued delivery.** Outbound eTIMS operations (item registration, sales, credit notes, stock movements) are written to a durable sync queue and executed by dedicated handlers, so a transient KRA or network failure does not lose a transaction — it is retried until KRA acknowledges it.
- **Idempotent receipt attach-back.** When KRA accepts a sale, the signed receipt (PDF + QR) is fetched once and attached to the originating invoice in the accounting platform, keyed to a unique compliance document ID so re-delivery cannot duplicate it.

3. eTIMS OSCU Integration Approach

sync2books integrates as the **Trader Invoicing System (TIS)** in KRA's OSCU model: the compliance service is the software component that talks to KRA's hosted OSCU, on behalf of each onboarded business, using that business's own KRA PIN and branch credentials.

3.1 Why OSCU

OSCU is hosted by KRA, so no signing hardware or client-side JAR needs to be distributed to each business — sync2books' backend calls KRA's API directly for every connected branch. This matches sync2books' model of serving many small and medium businesses from a single, centrally operated platform.

3.2 Certification and credentials

sync2books applies for certification as a third-party integrator via the eTIMS Bio Data Form. Once certified, KRA issues an Integration Token used on the eTIMS Service Request form. Per-branch operational credentials (branch office ID and communication key) are separate: they are returned by KRA's own device-initialization endpoint once a branch's Service Request is approved, and are stored encrypted against that branch's record in the compliance service.

4. Integration Sequence

Implementation follows the mandatory order defined in the OSCU/VSCU Step-by-Step Guide and Technical Specification v2.0. Each stage below maps to a concrete module in the sync2books platform.

#	Stage	sync2books component	eTIMS OSCU operation
1	Branch provisioning & initialization	Provisioning service creates a compliance tenant and per-branch OSCU connection, stores KRA PIN / branch office ID / device serial	Device initialization → communication key issued
2	Basic data	Compliance service	Get code list, item classification, PIN info, branch list, notices
3	Branch information	Provisioning service	Send head office / branch info and branch user accounts

#	Stage	sync2books component	eTIMS OSCU operation
4	Item (catalogue) management	Operational service — catalogue sync	Send / get item information
5	Sales & credit notes	Operational service — sale report, express credit note	Send sales transaction + sales invoice; credit note / cancel
6	Stock management	Operational service — stock transfer / adjust	Send stock in/out and stock inventory
7	Purchases	Operational service (extension)	Get purchase transactions; send confirmation
8	Receipt attach-back	Receipt attachment handler	Fetch signed receipt (PDF + QR) and attach to the originating invoice

4.1 Outbound transaction lifecycle

Every accounting-platform transaction that must reach KRA follows the same reliable pipeline:

- An invoice, credit note, item, or stock change is created or updated in the connected ERP and synced into sync2books' data model.
- The relevant eTIMS operation (e.g. sale report, catalogue sync, stock adjustment) is enqueued as a sync item with its target OSCU method and payload.
- The sync handler dequeues the item, calls the compliance service, which signs and submits it to KRA OSCU.
- The compliance service returns — synchronously for direct calls, or via an asynchronous outcome callback for longer-running submissions — KRA's acceptance, assigned receipt/invoice number, and signature.
- On acceptance, the signed receipt is retrieved and attached back to the invoice in the source accounting platform; on rejection, the item is marked failed with KRA's error detail and is available for retry.

5. Technology Stack

Layer	Technology
Backend framework	NestJS 11 (TypeScript) across the core sync engine and the compliance service
Persistence	MySQL (TypeORM), with versioned SQL migrations
Caching / rate limiting	Redis
Inter-service communication	HTTPS, bearer-token authenticated internal API between the core sync engine and the compliance service
ERP connectivity	OAuth2 (QuickBooks Online, MS Dynamics); native API integration (Odoo)
Partner / external API auth	API-key authentication with optional HMAC-SHA256 request signing
Dashboard authentication	JWT (access/refresh), bcrypt password hashing
Receipt generation	Server-side PDF generation with embedded QR code for each KRA-signed receipt
Frontends	React and Next.js single-page applications for tenant onboarding and compliance monitoring

6. Security and Credential Management

- **Credential custody.** KRA PIN, branch office ID, device serial and communication key are stored per branch in the compliance service only, never duplicated into the core sync engine or exposed to ERP connectors or frontends.
- **Encryption in transit.** All calls to KRA eTIMS OSCU, between internal services, and to ERP platforms use TLS/HTTPS.
- **Encryption at rest.** Sensitive connection fields (communication key, device serial) are encrypted at the database layer.
- **Service-to-service auth.** The core sync engine authenticates to the compliance service with a scoped bearer token; external/partner API access uses API keys with optional HMAC request signing to prevent tampering and replay.
- **Least privilege.** Each connected business's credentials are scoped to that business's own branch(es) only; there is no shared or pooled OSCU credential across tenants.
- **Auditability.** Every outbound eTIMS operation and its KRA response is persisted as a sync item with status, timestamps and error detail, giving a full audit trail per transaction.

7. Environments

Environment	eTIMS OSCU base URL	Purpose
Sandbox	https://etims-api-sbx.kra.go.ke	Development, integration testing and KRA certification review
Production	https://etims-api.kra.go.ke	Live submissions for onboarded, certified branches

Each branch connection is explicitly tagged with its environment; the compliance service selects the corresponding OSCU base URL and prevents sandbox-provisioned branches from submitting to production and vice versa.

8. Reliability and Error Handling

- **Durable queue.** Outbound eTIMS operations are persisted before submission, so a service restart or network interruption does not lose a transaction.
- **Automatic retry.** Failed submissions (network errors, KRA timeouts) are retried; partial failures reported inside a 2xx KRA response (e.g. a catalogue batch with mixed item-level results) are detected and the affected items re-queued individually.
- **Idempotency.** Receipt attach-back is keyed to a unique compliance document ID, so repeated delivery of the same outcome cannot create duplicate receipts on an invoice.
- **Status visibility.** Branch connection status (pending / connected / failed) and per-transaction sync status are exposed to the business through the compliance dashboard.

9. Glossary

Term	Meaning
TIS	Trader Invoicing System — the invoicing/business software that connects to eTIMS (sync2books).
OSCU	Online Sales Control Unit — KRA-hosted eTIMS integration path used by sync2books.
VSCU	Virtual Sales Control Unit — an alternative, client-hosted eTIMS integration path (not used by sync2books).
Communication key	Credential issued by KRA at device initialization, used to authenticate subsequent OSCU calls for a branch.
bhfid	KRA branch office identifier assigned during OSCU onboarding.
Integration Token	Token issued by KRA to a certified integrator, used on the eTIMS Service Request form.

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